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Course: Introduction to Big Data Analytics | Group A
Lab: Week 1 — Python Foundations
Institution: AUCA

WEEK 1 PYTHON FOUNDATIONS LAB REPORT

Introduction

This week's lab introduced the foundations of Python programming and showed how basic Python concepts can be used to work with data. I completed the exercises using Google Colab and organized each exercise in a separate code cell as required. The lab covered variables and data types, strings, lists, dictionaries, and basic analysis of a dataset containing 20 AUCA students.

What I Built and Learned

In Part 1, I created variables using different Python data types, including strings, integers, floats, and Boolean values. I also practiced type conversion by changing a number stored as text into an integer and using it in a calculation. I learned that "3" + "4" produces "34" because both values are strings, so Python joins them instead of performing numerical addition.

In Part 2, I worked with strings and learned how to use f-strings to combine variables with text. I also practiced converting text to uppercase, finding the length of a string, and using slicing to select specific characters.

In Part 3, I worked with lists containing quiz scores. I learned how to access elements, count values, calculate an average, and find the highest and lowest scores. I also modified the list by adding a new score and replacing an existing value.

In Part 4, I created a dictionary containing personal information using labeled keys such as name, age, program, and district. I practiced accessing, updating, and adding information to a dictionary.

In Part 5, I analyzed a dataset containing 20 AUCA students. I calculated the number of students, class average, highest and lowest scores, and score range. I also inspected individual records, counted students by district, and identified the student with the highest score.

What Surprised Me

One thing that surprised me was how easily Python can turn information from a dataset into something that can be analyzed. The student records looked like simple pieces of information, but by using lists, dictionaries, and Python functions, I could quickly calculate statistics and identify specific students. I was also surprised that putting a number inside quotation marks changes how Python interprets it and therefore changes the result of an operation.

Conclusion

Overall, this lab gave me a practical introduction to Python and basic data analysis. I learned how different Python data structures work and how they can be combined to analyze structured data. These foundations will be useful for understanding more advanced Big Data Analytics concepts in the coming weeks.